Tanmay Rawal

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Summary

ML/AI Engineer specializing in production machine learning, computer vision, edge inference, and GenAI systems. Built an **XGBoost electricity-demand forecasting pipeline at NTPC achieving ∼0.92% MAPE** and deployed quantized CNNs on PYNQ-ZU FPGA with up to **13.9× CPU acceleration and 98.4 FPS**. **Published at IHCI 2025 and accepted at IEEE TENCON 2026.**

Education

The LNM Institute of Information Technology — **Aug 2023 – May 2028**
B.Tech-M.Tech Integrated Dual Degree, ECE | ***CGPA: 7.65*** — *Jaipur, Rajasthan*

Publications

- **INSIGHT-BRAIN: Interpretable Neural Systems Using Grad-CAM, SHAP, and LIME in Human-Centered Brain Tumor Imaging** — *IHCI 2025* — **Published**
- **DPU-Accelerated Alzheimer's Disease Staging from Brain MRI Using Quantized ResNet50 on PYNQ-ZU** — *IEEE TENCON 2026* — **Accepted**

Experience

NTPC Ltd. (IT Department) — **Jun 2026 – Present**
ML Intern — Electricity Demand Forecasting GitHub — *Noida, India | Onsite*

- Developed an **XGBoost forecasting pipeline** across **4 horizons (5-min to 24-hr)** covering all Indian states & UTs, trained on **340K+ grid-telemetry records** (Jan 2023–Mar 2026) to support power-trading bid decisions on the Indian Energy Exchange.
- Engineered **21 predictive features** spanning calendar effects, live weather via Open-Meteo API, autoregressive lags (24h/7d), and historical baselines, achieving ∼**0.92% MAPE** on the 5-minute model against an industry threshold of <2%.
- Built a **Django backend** with APScheduler-based automated data ingestion and a real-time forecasting dashboard; optimized the **SQL + NumPy** pipeline for a **12× speedup**, generating a 288-block forecast in <**2.5 seconds**.

LUSIP, LNMIIT — **May 2025 – Aug 2025**
Research Intern — Brain Tumor MRI Classification GitHub — *Jaipur, Rajasthan*

- Developed a **weighted ensemble** of EfficientNetB0, InceptionV3, and Xception over **7,421 MRI scans**, achieving **98.8% test accuracy** and improving performance by **2–4% over individual-model baselines**; applied **Grad-CAM, LIME, and SHAP** for explainability.
- INT8-quantized the model using **Vitis AI**, compiled it for the **DPUCZDX8G DPU**, and deployed inference on a **PYNQ-ZU FPGA**, retaining accuracy within **0.3% of FP32** while enabling real-time edge inference without GPU or cloud dependency.

Projects

Alzheimer's MRI Classification — INT8 DPU Inference on PYNQ-ZU — GitHub

- Trained **ResNet50** on **9,488 OASIS MRI scans** using Focal Loss, achieving **96.42% accuracy and 0.9970 AUC**; performed **INT8 post-training quantization** with Vitis AI and compiled the model to `.xmodel` for **DPUCZDX8G_ISA1_B4096**.
- Achieved **10.15 ms inference latency, 98.4 FPS, and 13.9× CPU speedup** across **423 DPU operations** with **zero CPU fallback**; reduced model size from **92.3 MB FP32 to 24.49 MB INT8**.

ResolveAI — Multi-Agent RAG System with MCP Integration — GitHub

- Built a **4-agent sequential workflow** using CrewAI + Gemini: Triage → Policy Researcher with **FAISS retrieval** → Resolution Architect → Compliance Guard, with automated hallucination, citation, and policy-compliance checks.
- Evaluated the pipeline on a **23-ticket benchmark**, achieving **100% citation coverage and 100% compliance pass rate**; exposed the system as an **MCP server using FastMCP** for Claude Desktop with an interactive Streamlit interface.

Technical Skills

Languages	Python, C++, SQL, CUDA, VHDL
ML / AI	PyTorch, TensorFlow, Scikit-learn, XGBoost, Deep Learning, Computer Vision, Time-Series Forecasting, XAI, Feature Engineering
GenAI / LLM	RAG, LangChain, CrewAI, FAISS, Hugging Face, Sentence Transformers, Multi-Agent Systems, MCP
Deployment / Edge	Docker, Linux, Git, ONNX, TensorRT, Vitis AI, PYNQ-ZU, FPGA Acceleration, INT8 Quantization
Libraries	NumPy, Pandas, OpenCV, Streamlit

Achievements

- **Rank 1** in B.Tech-M.Tech Integrated Dual Degree Program, LNMIIT Jaipur.
- **Ranked 54th among 1,200+ teams globally** (**Top 4.5%**) in the **Nokia FPGA Hackathon**; selected as a Top 100 Finalist.